OPERATING SYSTEMS

UE25CS242A

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File Management

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OPERATING SYSTEMS

Access Control, Revocation of Access Rights

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Slides Credits for all PPTs of this course

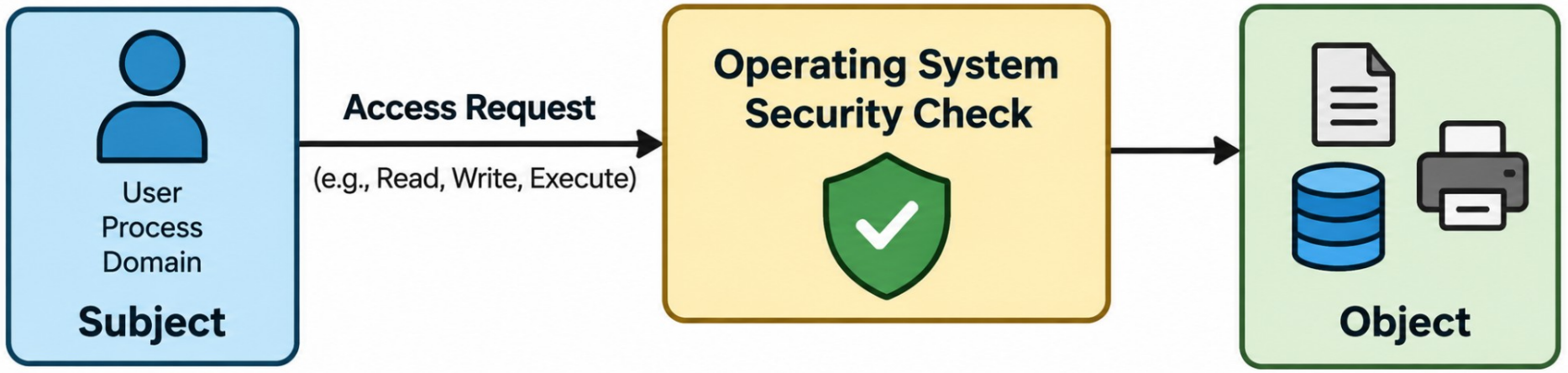


- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau

Access Control

- Access control determines who can access a resource and what operations they are allowed to perform.
- The operating system uses protection mechanisms to control access to files, devices, and other objects.
- Access rights may be assigned to users, domains, or roles.



Revocation of Access Rights



- Revocation means removing previously granted access rights.
- It is required when access permissions need to be changed or withdrawn.
- Revocation is important in dynamic protection systems, where access rights can change over time.

Example

Before:

Faculty → File F1 → Read + Write

After Revocation:

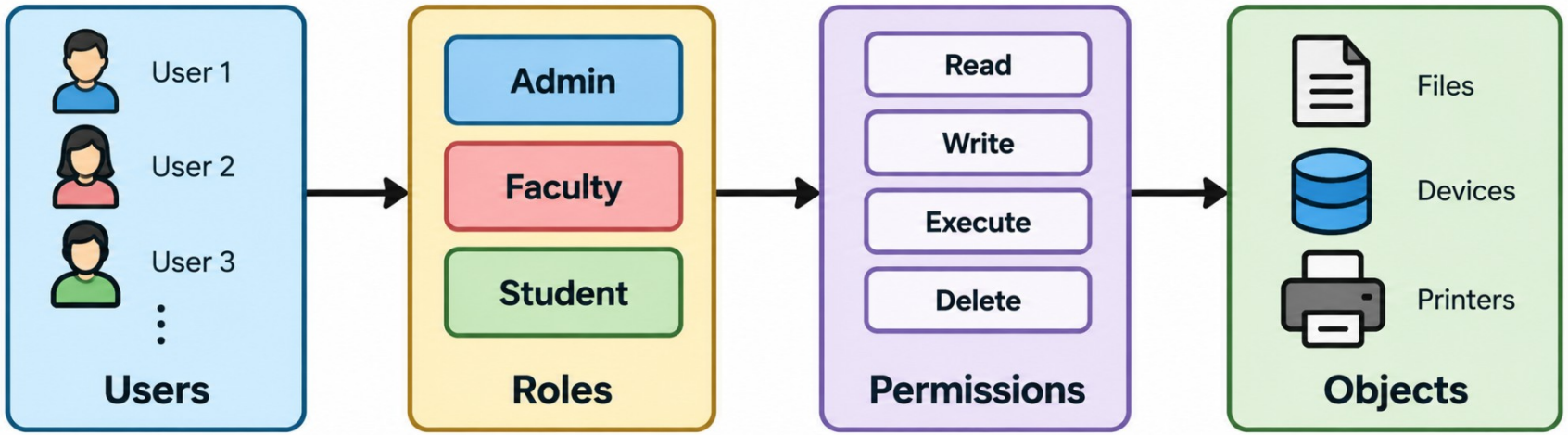
Faculty → File F1 → Read

→ The **Write** permission has been revoked.

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Role-Based Access Control (RBAC)

- In RBAC, permissions are assigned to roles, rather than directly to individual users.
- A role represents a user's job or function in the system.
- Users are assigned one or more appropriate roles.



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Why Revocation of Access Rights is Needed



Revocation

- Revocation means removing previously granted access rights from a user or domain.
- It is necessary because access requirements can change over time.
- Revocation is an important part of a dynamic protection system.

Why is Revocation Required?

- A user's role or responsibility changes.
- A user no longer needs access to a resource.
- A resource becomes more restricted.
- Temporary access has expired.
- An administrator needs to withdraw previously granted permissions.

Simple Example

Before: D1 → File F1 → Read + Write

After Revocation: D1 → File F1 → Read : The **Write** right has been removed, while **Read** is retained.

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Types of Revocation



Revocation can be classified based on when, how much, and for whom access is removed.

1. Immediate vs. Delayed

- **Immediate:** Access is removed immediately.
- **Delayed:** Revocation takes effect after a specified time or condition.

2. Selective vs. General

- **Selective:** Remove access from a specific user/domain.
- **General:** Remove access from all users/domains having that right.

3. Partial vs. Total

- **Partial:** Remove some rights while keeping others.
- **Total:** Remove all access rights to the object.

4. Temporary vs. Permanent

- **Temporary:** Access is removed for a limited period or until a condition is met.
- **Permanent:** Access is removed and does not automatically return.

Revocation Using ACL and Capability List



Revocation with Access Control List (ACL)

- In an ACL, each object maintains a list of domains/users and their access rights.
- To revoke access, the OS can remove or modify the corresponding entry.
- Therefore, revocation using ACL is relatively straightforward.

Revocation with Capability List

- In a Capability List, access rights are associated with domains.
- Capabilities may be distributed among different domains.
- Therefore, revoking a particular capability can be more difficult.

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Revocation Techniques



When capabilities are distributed among domains, different techniques can be used to revoke access.

1. Reacquisition

- Capabilities have a limited validity.
- When they expire, the user/domain must request access again.
- If access is revoked, the request is denied.

2. Back-Pointers

- The system maintains pointers to all copies of a capability.
- During revocation, these pointers are used to locate and modify/remove the capabilities.

3. Indirection

- Access is managed through an intermediate/global table.
- To revoke access, the corresponding entry in the table is deleted or invalidated.

4. Keys

- A capability is associated with a key.
- Changing or invalidating the key makes the old capability unusable.

Access Matrix



- Protection model represented as a **matrix**
- **Rows → Domains**
- **Columns → Objects**
- Each cell contains **access rights**

access(i, j):

- Operations allowed on object **Oj** by domain **Di**

Access Matrix example



- Domains: **D1, D2, D3, D4**
- Objects: **F1, F2, F3, Printer**

Key Points:

- **D1** → Read F1, F3
- **D4** → Same as D1 + Write F1, F3
- **D2** → Access Printer only

object / domain	F_1	F_2	F_3	printer
D_1	read		read	
D_2				print
D_3		read	execute	
D_4	read write		read write	

Access Matrix



Ensures:

- Only **allowed access**
- Strict control of resources

OS decides:

- Which domain a process runs in
- What rights are given

Domain Switching

- Domain is treated as an **object**
- Switching = performing **switch operation**

Rule:

- Process can switch from **Di → Dj**
  only if 'switch' right exists

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Access Matrix – Domain switching



- D2 → can switch to D3, D4
- D4 → can switch to D1
- D1 → can switch to D2

Additional Rights:

- **copy** → duplicate access rights
- **owner** → full control
- **control** → modify rights

object / domain	F_1	F_2	F_3	laser printer	D_1	D_2	D_3	D_4
D_1	read		read			switch		
D_2				print			switch	switch
D_3		read	execute					
D_4	read write		read write		switch			

Implementation of the Access Matrix – Global table

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- Represented as: **<domain, object, rights>**
- System searches table to check permission

Drawbacks:

- Large size → not memory efficient
- Slow lookup (search required)
- Difficult to manage grouping

Implementation of the Access Matrix – Access list for objects



- Each object has a list:
  - **(domain, rights)**

Advantages:

- Easy to check **who can access object**
- Supports default permissions

Limitation:

- Hard to know what a domain can access

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Role based access control

- Access permissions are assigned to **roles**, not directly to users
- A **role** represents a job/function (e.g., Admin, Faculty, Student)

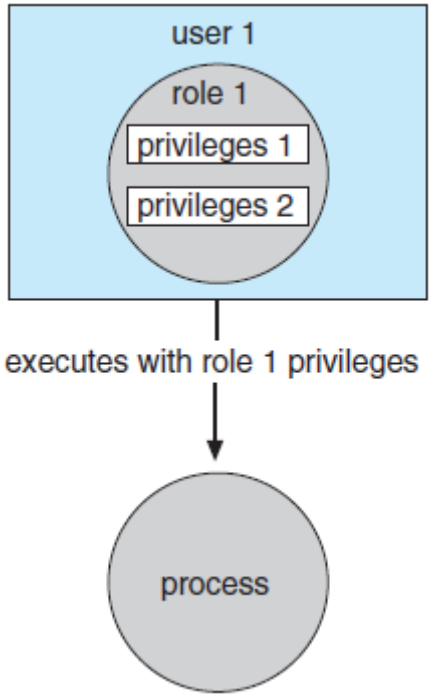
How it works:

- **Users → assigned to roles**
- **Roles → assigned permissions**
- Users gain access **through roles**

Example:

- **Admin** → read, write, delete
- **Faculty** → read, write
- **Student** → read only

If a user is assigned “Faculty” role → gets those permissions



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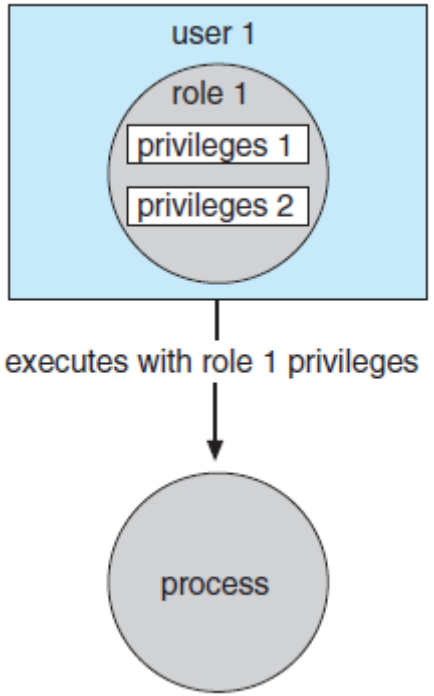
Role based access control

Advantages:

- Reduces complexity in access management
- Easy to update permissions (change role, not every user)
- Improves security (no direct permission assignment)
- Supports large systems efficiently

Limitation:

- Requires proper role design
- Too many roles → becomes complex



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Revocation of access rights



- **Revocation** = Removing access rights from users/domains
- Needed in **dynamic protection systems**

Types of Revocation:

- **Immediate / Delayed**
  - Instant removal vs later
- **Selective / General**
  - Specific users vs all users
- **Partial / Total**
  - Some rights vs all rights
- **Temporary / Permanent**
  - Can be regained vs removed forever

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Revocation of access rights



Access List (ACL):

- Easy to revoke:
  - Simply **remove entry from list**
- Supports:
  - All types (selective, partial, etc.)

Capability List:

- Difficult to revoke:
  - Capabilities are **distributed across system**

Revocation of access rights



Revocation Techniques:

- **Reacquisition**
  - Capabilities expire → must request again
- **Back-pointers**
  - Track all capabilities → modify them
- **Indirection**
  - Use global table → delete entry
- **Keys**
  - Change key → invalidate all old access



THANK YOU

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